



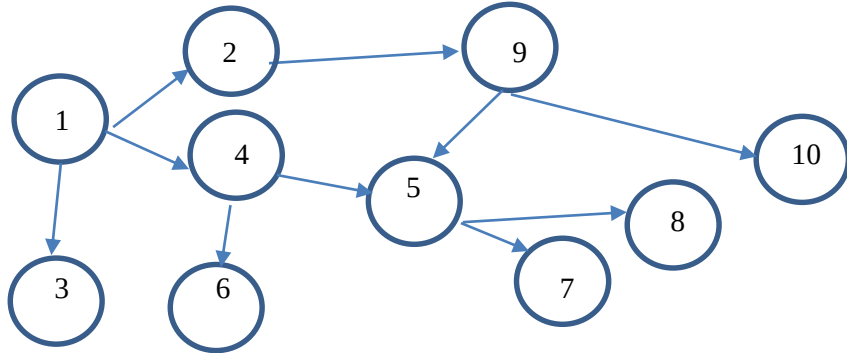
**NATIONAL INSTITUTE OF TECHNOLOGY
ROURKELA – 769 008**

Department	Computer Science and Engineering
Examination	End Semester – 2024-25
Program	B. Tech / M. Tech / Ph. D
Session	Spring
Course code & Course Name	CS6404 - Software Testing
Marks	50
Duration	3 Hours

This question paper contains 2 pages

Answer all questions.

Q. No.	Questions	Marks	CO*	BL**		
1.	Consider a program for the determination of division of a student based on the marks in 3 subjects. Its input is a triplet of positive numbers (say, mark1, mark2, mark 3) and the values are from interval [0 – 100]. The division is calculated as follows:	5	CO-1	BL-3		
	Sl. No.				Marks obtained (Average)	Division
	1				75 - 100	First division with Hons.
	2				60 - 74	First division
	3				50 - 59	Second division
	4				40 - 49	Third Division
	5				0 - 39	Fail
	Total marks obtained are the average of marks obtained in the three subjects. The output may have any one of the above divisions. Design test cases for the above problem using equivalence class partitioning method.					
2.	Consider the program for classification of a triangle. Its input is a triplet of positive numbers (say, a, b, c) and they satisfy the constraint 0<input parameters<=100. The triangle is classified according to the following rules: Right angled triangle: $c^2=a^2+b^2$ or $a^2=b^2+c^2$ or $b^2=c^2+a^2$ Obtuse angled triangle: $c^2>a^2+b^2$ or $a^2>b^2+c^2$ or $b^2>c^2+a^2$ Acute angled triangle: $c^2<a^2+b^2$ or $a^2<b^2+c^2$ or $b^2<c^2+a^2$ The output may have any one of the above triangles. Design test cases using decision table based testing.	5	CO-1	BL-3		
3.	Consider the following program segment: main() { int a, b, c, x=0, y=0; printf(“Enter 3 numbers”); scanf(“%d%d%d”, &a, &b,&c); if ((a>b)&&(a>c)){ x=a*a+b*b; } if (b>c){ y=a*a-b*b; } printf(“x=%d y=%d”, x, y); } Draw the CFG for the above function. Determine the cyclomatic complexity. List all the independent paths. Design test cases for all independent paths.	5	CO-2	BL-3		

4.	Consider the above program given in Q. 3. Find out 4 possible mutants in the program and check how many of them are killed by creating a set of test data. Compute the mutation score. Add new test data, if required, to improve the mutation score.	5	CO-2	BL-3																								
5.	What are the main differences among alpha, beta and acceptance testing techniques? How are the test cases designed for these tests? For each of the three tests, explain who carries out the test, when is the test carried out, and the objectives of the test.	5	CO-3	BL-3																								
6.	Perform pair-wise integration testing using the following call graph. Show all the intermediate steps. Calculate the testing effort required for pair-wise integration testing for the following call graph. 	5	CO-3	BL-3																								
7.	What do you mean by regression test suite minimization? Explain the regression test suite minimization technique along with a suitable diagram and the necessary steps.	5	CO-4	BL-2																								
8.	Consider the details of the following Guest House Management System. <table><tr><td>Sl. No.</td><td>Functionality</td><td>Function ID in SRS</td><td>Test Cases</td></tr><tr><td>1</td><td>Login the System</td><td>F3.4</td><td>T1</td></tr><tr><td>2</td><td>View Room Availability Details</td><td>F3.5</td><td>T2</td></tr><tr><td>3</td><td>Book a Room</td><td>F3.6</td><td>T3</td></tr><tr><td>4</td><td>Manage Rooms</td><td>F3.7</td><td>T4</td></tr><tr><td>5</td><td>Exit the System</td><td>F3.8</td><td>T5</td></tr></table> Develop the test specification and test log for ‘View Room Availability Details’ functionality.	Sl. No.	Functionality	Function ID in SRS	Test Cases	1	Login the System	F3.4	T1	2	View Room Availability Details	F3.5	T2	3	Book a Room	F3.6	T3	4	Manage Rooms	F3.7	T4	5	Exit the System	F3.8	T5	5	CO-4	BL-4
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5	Exit the System	F3.8	T5																									
9.	Apply Extended Use Case Test Pattern to “Withdraw Cash” use case of an ATM System and generate the test cases. OR What is random class testing? Generate 4 test cases for the <i>Book Class</i> in a Library Information System using random class testing. You may assume your own operations / methods for the <i>Book Class</i> .	5	CO-5	BL-4																								
10.	How software is developed using scrum? Briefly explain the testing process in different phases of scrum. OR Suppose you are given the task of developing a web application for Guest House Management at NIT Rourkela. Explain the following types of web testing that you need to perform for the above web application. (i) Usability testing (ii) Navigation testing	5	CO-5	BL-2																								